MILITARY SPECIFICATION

TEST SET, ELECTRONIC CIRCUIT PLUG-IN UNIT AN/ASM-152

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The test equipment covered by this specification shall be for shop use and shall be capable of checking the plug-in modules of the Weapons Release Computer CP-741 (V)/A and the associated Pilot's Control Module (Weapon Control Panel).

1.2 Classification - The Test Equipment covered by this specification shall consist of the following items

Type Designation	Applicable Paragraph
TS-2227A/ASM-152 Test Set, Electronic Circuit Plug-In Unit	3.6.1
CY-4629/ASM-152 Case, Test Set	3.6.2
CX-10159/ASM-150 Cable Assembly, Electrical, Branched	3.6.3
CX-11043/ASM-152 Cable Assembly, Special Purpose, Electrical	3.6.4
J-2556 ASM-152 Distribution Box	3.6.5
63A25D24 Tool, Module Puller and Disconnect	3.6.6
58A5A294 Wrench, Hex	3.6.7

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

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SPECIFICATIONS

Military

MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for

STANDARDS

Military

MIL-STD-454	Standard Requirements for Electronic Equipment
MIL-STD-470	Maintainability Program Requirements for Systems and Equipments
MIL-STD-471	Maintainability Demonstrations
MIL-HDBK-472	Military Standardization Handbook, Maintainability Prediction
MIL-STD-781	Test Level and Accept/Reject Criteria for Reliability of Non-expendable Electronic Equipment
MIL-W-27076	Workmanship Standards for Ground Electronic and Associated Equipment

BULLETINS

ANA Bulletin No. 400	Air Force-Navy Aeronautical Bulletin Electronic Equipment Aircraft and Guided Missiles, Applicable Documents

DRAWINGS

Naval Air Systems Command

List of Drawings DL 63A24E1	All drawing numbers listed in this specification are a part of DL 63A24E1 except the following:

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64A6A100 (NASC)	Quality Assurance Information Requirements
64A7A100 (NASC)	Quality Assurance Information Requirements

2.1.1 Availability of Documents -

(1) Contractors requesting documents required in connection with specific procurement functions shall, on the application, refer to the documents by both title and number.

(2) Copies of this specification and specifications, standards, and publications referenced herein and in DL 63A24E1 may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

(3) Copies of DL 63A24E1 and the drawings listed therein may be obtained upon application to the Commanding Officer, Naval Air Technical Services Facility, 700 Robbins Avenue, Philadelphia, Pennsylvania 19111.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - Parts and materials shall be as specified in DL 63A24E1. Any deviation by the contractor from the use of these specified parts and materials necessarily places both the burden of proof of the acceptability of the change, and the cost associated therewith, upon the contractor. Approval for deviations must be requested from the procuring activity by the contractor and granted by the procuring activity prior to submission of equipments for preproduction tests. A request for approval shall include, as supporting evidence, the data specified in 3.3. Approval of parts and material deviations shall not be construed by the contractor to constitute a waiver of any other requirement specified in this specification or in the applicable drawings of DL 63A24E1.

3.2.1 Quality of Parts Requirements - The contractor shall be responsible for assuring that all electrical, electromechanical and electronic parts meet the requirements of the "Preferred" or "Qualified" category as defined herein (see 6.4) Any information obtained from a component parts vendor by any other company or government activity for placing the parts in the "Preferred" or "Qualified" category shall not be applicable (exceptions to this are parts manufactured to conventional "MIL", "MS", or "AN" Specifications having Qualification Approval (QPL) requirements). The lots of any parts meeting only the requirements of the "Preliminary" category, as defined herein (see 6.4) must be qualified individually by passing such tests as necessary to determine conformance to critical requirements. Data required to classify a part shall be made available to the procuring activity upon request.

3.3 Supplemental Requirements - The following supplemental requirements for the selection of parts and materials are applicable only in the event that the contractor proposes to alter the equipment design delineated by the drawings of DL 63A24E1.

3.3.1 Conformance to MIL-T-21200 - Parts and materials selected for use in the equipment shall conform to the requirements of Specification MIL-T-21200 Parts shall be selected from those parts included in ANA Bulletin No 400, unless the parts listed therein will not perform adequately in the applications and environments specified herein.

3.3.2 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained by the contractor as outlined in Specification MIL-T-21200 (see 6.3).

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3.4 Design and Construction - The equipment shall be manufactured in accordance with the drawings and subsidiary specifications listed in DL 63A24E1, except insofar as errors of omission or commission on the drawings may be incompatible with the requirements of this and related specifications. In any case, it is a condition of final acceptance that the equipment meet all the performance requirements specified herein and such acceptance test documents that shall be applicable. Proposed changes, either to correct errors in the design disclosure or to effect improvements, shall be specifically approved in writing by the procuring activity unless otherwise specified.

3.4.1 Reliability -

3.4.1.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently, for a period of at least 125 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.4.1.2 Operating Life - The equipment shall have a minimum total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment, and the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency (see 6.3).

3.4.1.3 Reliability in Mean-Time-Between-Failure - The equipment shall have a minimum of 300 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.4.2 Interchangeability - If the contractor proposes to alter the equipment design as described in 3.3, the drawings listed on DL 63A24E1 shall be the standard for measuring interchangeability of all replaceable assemblies, subassemblies, and parts

3.4.3 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of MIL-I-6181.

3.4.4 Mean-Time-To-Repair (MTTR) - The mean-time-to-repair (MTTR) of the equipment shall be not greater than 2.5 hours when replacing circuit modules, plug-ins, assemblies and parts that are intended for replacement at the organizational level. Only where design changes shall cause significant deviations from procuring activity drawings, a prediction of MTTR shall be made in accordance with MIL-STD-470 and MIL-HDBK-472, PROCEDURE II, and shall be demonstrated in accordance with 4.2.2

3.4.5 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (30 ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient
Input Power	115 ± 11.5 V AC 400 ± 20 cps and 27.0 ± 2.0 V DC

3.4.6 Service Conditions - The equipment shall operate within the specified tolerance limits of 3.6 under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class 3 equipment except as specified below.

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3.4.6.1 Shock - In addition to the shock requirements of MIL-T-21200 the test equipment, when stored in its combination case, shall operate as specified in 3.5 after subjection to two half sine wave, 70g shocks, each of seven milliseconds duration applied upward along the vertical axis of the test equipment case. Physical damage to the equipment is permissible provided the operational requirements of 3.5 are maintained.

3.4.6.2 Vibration Test - The vibration test shall be performed with the test equipment resting on its four feet and the front panel in a horizontal position.

3.4.7 Warm-Up Time - The equipment shall require a maximum of 15 seconds warm-up time above +10°C and a maximum of 2 minutes below +10° Centigrade.

3.4.8 Primary Input Power Requirements - The power required shall be of the following types and shall not exceed the specified amounts:

AC Power, Single Phase, 115 V, 400 cps, 70 VA

DC Power, 28 V, 28 Watts

3.5 Performance - The performance of the equipment shall be as set forth below. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variation will be specified

3.6 Detail Requirements -

(1) Normal Operating Position - The normal operating position of the subject test set shall be with the front panel in a near-vertical position

(2) Measuring Equipment - Electrical requirements specified herein shall be measured with equipment of the following requirements as a minimum:

DC voltage measurement - Measurement to 5 significant figures with an accuracy of ± 0.01% of reading, ± 1 digit in the 5th significant figure.

AC voltage measurement - Measurement to 4 significant figures with an accuracy of ± 0.1% of reading, ± 2 digits in the 4th significant figure.

Resistance measurement - Measurement to 5 significant figures with an accuracy as follows

0 to 999.99 ohms ± (0.01% of reading ± 2 digits)

1.0000 K to 99.999 K ± (0.01% of reading ± 1 digit)

100 K to 999.99 K ± (0.03% of reading ± 1 digit)

Voltage output - Adjustable to the nearest 5th significant figure with an accuracy of ± 0.01% of setting or ± 100 μv, whichever is greater

Current output - Adjustable to the nearest 5th significant figure with an accuracy of ± 0.02% of setting or ± 0.002% of range, whichever is greater.

Time measurement - Measurement to 4 significant figures with an accuracy of ± 0.01% of reading.

Resistance applied external to test set - Accuracy of 0.03% unless otherwise specified.

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3.6.1 Test Set, Electronic Circuit Plug-In Unit TS-2227A/ASM-152 - The test set shall be manufactured in accordance with the applicable drawings of DL 63A24E1 and shall meet the following requirements:

3.6.1.1 Function - The test set shall test the performance of the plug-in modules of Weapons Release Computer CP-741 (V)/A, and a Pilot's Control Module (Weapon Control Panel) in the shop.

3.6.1.2 Electrical Requirements - The test set shall meet the following electrical requirements:

3.6.1.2.1 Adjustment - The 115 VAC, 400 cycles entering the test set through the POWER connector J101 shall be adjusted as follows:

(1) Set the MODULE SELECT switch to the LINE position.

(2) Adjust the AC VOLTAGE ADJUST autotransformer for a GOOD indication on the QUALITY meter.

(3) Depress the MAX SENS pushbutton and adjust the AC VOLTAGE ADJUST autotransformer until the QUALITY meter indicates on the 0 mark.

In order to meet the requirements specified herein, it may be necessary to perform the above adjustment more than once during a performance test. The adjustment should change only when the 115 VAC, 400-cycle voltage entering the test set changes.

3.6.1.2.2 Test Circuit Requirements - The test circuit shall test the internal power supply with the MODULE SELECT switch in the "-10" (3rd position) and "+10" (4th position) positions. The test circuit shall cause the QUALITY meter to indicate as specified in table I with the stated input voltages applied. The "QUALITY Meter 0 Mark" column is for reference and shall not be a test requirement.

TABLE I

MODULE SELECT Switch Position	Input Voltage		Input Voltage* (volts d-c)		
			QUALITY Meter 0 Mark	QUALITY Meter Border Marks Between the GOOD Region and Two BAD Regions	
	From	To		Left Mark	Right Mark
-10 +10	J102-8(-) J102-10(+)	J102-6 J102-5	-10.000 +10.000	-10.015 +10.015	-9.985 +9.985

*All input voltages may be adjusted within a tolerance of ± 0.006 volt of the specified voltage to obtain the desired QUALITY meter indication. Although the input voltage shall be applied where indicated in table I the value of this voltage shall be measured at the points indicated in table II.

3.6.1.2.3 Power Supply Module Test Circuit - The power supply module test circuit shall meet the following requirements with the MODULE SELECT switch in "PS" position (5th position):

3.6.1.2.3.1 QUALITY Meter Indications - The power supply module test circuit shall

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cause the QUALITY meter to indicate as specified in table III with the applied input voltages. All input voltages may be adjusted within a tolerance of ± 0.006 volt d-c of the specified voltages. The "QUALITY Meter 0 Mark" column is for reference and shall not be a test requirement.

TABLE II

MODULE SELECT Switch Position	Input Voltage Measurement Points	
	From	To
-10	J102-7(-)	J102-12
+10	J102-9(+)	J102-11

TABLE III

POWER SUPPLY Switch Position	Input Voltage		Input Voltage* (volts d-c)		
	From	To	QUALITY Meter 0 Mark	QUALITY Meter Border Marks Between GOOD Region and Two BAD Regions	
				Right Mark	Left Mark
1	J201-10(+)	J201-5	+10.000	+9.985	+10.015
2	J201-10(+)	J201-5	+10.000	+9.985	+10 015
3	J201-8(-)	J201-6	-10.000	-9.985	-10.015
4	J201-8(-)	J201-6	-10.000	-9.985	-10.015

*Although the input voltage shall be applied where indicated, the value of this voltage shall be measured at the points indicated in table IV.

TABLE IV

POWER SUPPLY Switch Position	Input Voltage Measurement Points	
	From	To
1	J201-9(+)	J201-11
2	J201-9(+)	J201-11
3	J201-7(-)	J201-12
4	J201-7(-)	J201-12

3.6.1.2.3.2 10-Volt D-C Lamp Indications -

(1) The power supply module test circuit shall cause the 10-volt d-c lamp to illuminate when 10 volts d-c is applied between J201-10(+) and J201-5 with the POWER SUPPLY switch in positions 1 or 2.

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(2) The power supply module test circuit shall cause the 10-volt d-c lamp to illuminate when 10 volts d-c is applied between J201-8(-) and J201-6 with the POWER SUPPLY switch in positions 3 or 4.

3.6.1.2.3.3 Voltage Output - The power supply module test circuit shall provide 115 ± 0.8, 119.5 ± 1.6, and 107.5 ± 1.6 volts a-c output between pins J201-1 and J201-3 with the AC VOLTAGE SELECT switch set to NOM., HIGH, and LOW, respectively.

3.6.1.2.4 Warning Circuit Module Test Circuit - The warning circuit module test circuit shall meet the following requirements with the MODULE SELECT switch in WRN position (6th position).

3.6.1.2.4.1 Warning Circuit - The warning circuit test circuit shall meet the following requirements.

3.6.1.2.4.1.1 WARN Lamp Indications - The warning circuit test circuit shall illuminate the WARN lamp when a ground is provided to the J301 pins specified in table V. (When testing a good CP-741(V)/A Computer Warning Circuit Module the WARN lamp should illuminate only in WARNING switch positions 3, 6, 9 and 12.)

TABLE V

WARNING Switch Position	Ground Pin J301-
1	7
2	7
3	7
4	9
5	9
6	9
7	12
8	12
9	12
10	14
11	14
12	14

3.6.1.2.4.1.2 Voltage-Current Outputs - The warning circuit test circuit shall, as specified in table VI, provide the following voltage and current outputs across (voltage) or through (current) the specified loads connected between connector J301-2 (ground) and the specified J301 pins with the WARNING switch in the designated positions in table VI

3.6.1.2.4.1.3 Resistance Outputs - The warning circuit test circuit shall provide the the resistance outputs specified in table VII at connector J301 with the WARNING switch in any position 1 through 12.

3.6.1.2.4.2 Absolute Summing Magnetic Amplifier Test Circuit - The absolute summing magnetic amplifier test circuit shall meet the following requirements.

3.6.1.2.4.2.1 Resistance Outputs - The absolute summing magnetic amplifier test circuit shall, as specified in table VIII, provide the following resistance outputs at connector J301 with the WARNING switch in the designated positions in table VIII.

3.6.1.2.4.2.2 Output Test Currents - The absolute summing magnetic amplifier test circuit shall provide output test currents as specified in table IX through any load in the range of 185 to 1000 ohms connected between connector J301-15 and J301-2 (ground) with the WARNING switch in the designated positions in table IX.

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TABLE VI

WARNING Switch Position	Voltage Obtained at	Voltage or Current	Resistance Load Range (ohms)
1	J301-6	14.490 ± 0.062 v d-c	700 to 1000
2	J301-6	13.650 ± 0.052 v d-c	700 to 1000
3	J301-6	13.300 ± 0.042 v d-c	700 to 1000
4*	J301-8	16.000 ± 0.062 v d-c	850 to 1000
5*	J301-8	15.150 ± 0.052 v d-c	850 to 1000
6*	J301-8	14.700 ± 0.042 v d-c	850 to 1000
7*	J301-8	6.923 ± 0.041 v d-c	850 to 1000
8*	J301-8	6.300 ± 0.041 v d-c	850 to 1000
9*	J301-8	5.747 ± 0.036 v d-c	850 to 1000
10	J301-13	1.428 ± 0.031 ma	850 to 1000
11	J301-13	1.199 ± 0.031 ma	850 to 1000
12	J301-13	0.950 ± 0.031 ma	850 to 1000
all	J301-3	+10.000 ± 0 015 v d-c**	N/A
all	J301-4	-10.000 ± 0.015 v d-c**	N/A
all	J301-1	115 ± 0.8 v a-c (VOLTAGE SELECT switch set at NOM.)	N/A
all	J301-1	119 5 ± 1 6 v a-c (VOLTAGE SELECT switch set at HIGH)	N/A
all	J301-1	107.5 ± 1.6 v a-c (VOLTAGE SELECT switch set at LOW)	N/A
all	J301-21	27.0 ± 2.0 v d-c	N/A

*A 1000 ohm resistor shall be connected between J301-10 and J301-2 (ground).

**These voltages shall be brought through a nonshorting deck of the WARNING switch so that no voltage is applied to pins J301-3 or J301-4 between switch positions.

TABLE VII

Resistance Obtained		Resistance (ohms)
From	To	
J301-11	J301-3	901.5 ± 3.7
J301-20	J301-3	900 ± 2.1
J301-5	J301-2	0 to 0.3

3.6.1.2.4.2.3 QUALITY Meter Indications - The absolute summing magnetic amplifier test circuit shall cause the QUALITY meter to indicate as specified in table X with input currents applied to connector J301-35 and the WARNING switch in the designated position. A 1000 ± 50-ohm resistor shall be connected between connector J301-15 and J301-2. The "QUALITY Meter on 0 Mark" column in table X is for reference and shall not be a testing requirement. The tolerances on the values obtained in this paragraph shall be as determined in the next paragraph.

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TABLE VIII

WARNING Switch Position	Resistance Obtained		Resistance (ohms)
	From	To	
13	J301-36	J301-2	1569 ± 3.8
14	J301-36	J301-2	1026 ± 2.7
15	J301-36	J301-2	833.2 ± 2.1
all	J301-33	J301-2	0 to 0.3
all	J301-5	J301-2	0 to 0.3
all	J301-16	J301-17	0 to 0.3
all	J301-18	J301-34	0 to 0.3

TABLE IX

WARNING Switch Position	Output Test Current (ma)
13	0.735 ± 0.025
14	3.312 ± 0.030
15	7.360 ± 0.035

TABLE X

WARNING Switch Position	QUALITY Meter on 0 Mark	Input Current (ma) QUALITY Meter on Border Marks Between the GOOD Region and Two BAD Regions	
		Right Border Mark	Left Border Mark
13	1.000	1.080	0.920
14	4.500	4.565	4.435
15	10.000	10.080	9.920

3.6.1.2.4.2.4 Tolerances - There are no fixed tolerances on the values in table X. Instead, the total absolute summing magnetic amplifier test circuit error shall be determined by using the values obtained in 3.6.1.2.4.2.2 and the values obtained in 3.6.1.2.4.2.3. The total circuit error shall be computed as follows for each of the WARNING switch positions 13 through 15.

(1) The value obtained per 3.6.1.2.4.2.2 shall be I_{in}. Solve for I_o in the following equation:

$$I_o = 1.359\ I_{in}$$

(2) Determine the difference between I_o and the value in the "QUALITY Meter on 0 Mark" column of table X. If I_o is less than the value in table X, the difference value is negative. Let this difference value be "Error 1".

(3) Determine the difference between the value obtained per 3.6.1.2.4.2.3 for the right border mark and the value specified in the "Right Border Mark" column of table X. If the value obtained per 3.6.1.2.4.2.3 is

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less than the value specified in the "Right Border Mark" column of table X, the difference value is negative. Let this difference value be "Error 2".

(4) Determine the difference between the value obtained per 3.6.1.2.4.2.3 for the left border mark and the value specified in the "Left Border Mark" column of table X. If the value obtained per 3.6.1.2.4.2.3 is less than the value specified in the "Left Border Mark" column of table X, the difference value is negative. Let this difference value be "Error 3".

(5) Calculate "Error 4" by taking the algebraic difference between "Error 1" and "Error 2".

(6) Calculate "Error 5" by taking the algebraic difference between "Error 1" and "Error 3".

(7) The absolute value of both the "Error 4" and "Error 5" values shall be less than 40 microamperes.

3.6.1.2.4.3 Bootstrap Test Circuit - The bootstrap test circuit shall meet the following requirements:

3.6.1.2.4.3.1 QUALITY Meter Indications - The bootstrap test circuit shall cause the QUALITY meter to indicate as specified in table XI with the applied pairs of input currents. The input current values causing the QUALITY meter to indicate 0 are for reference and shall not be a test requirement

TABLE XI

WARNING Switch Position	Input Current No. 1		Input Current No. 2		QUALITY Meter Indication
	Input Current Applied To	Input Current (ma)	Input Current Applied To	Input Current (ma)*	
16	J301-37	1.13	J301-39	1.13 ± 0.02	0
16	J301-37	1.13	J301-39	0.93 ± 0.02	Border**
16	J301-37	1.13	J301-39	1.33 ± 0.02	Border**
17	J301-37	1.13	J301-41	1.13 ± 0.02	0
17	J301-37	1.13	J301-41	0.93 ± 0.02	Border**
17	J301-37	1.13	J301-41	1.33 ± 0.02	Border**
18	J301-37	1.13	J301-43	1.13 ± 0.02	0
18	J301-37	1.13	J301-43	0.93 ± 0.02	Border**
18	J301-37	1.13	J301-43	1.33 ± 0.02	Border**
19	J301-31	1.13	J301-45	1.13 ± 0.02	0
19	J301-31	1.13	J301-45	0.93 ± 0.02	Border**
19	J301-31	1.13	J301-45	1.33 ± 0.02	Border**
20	J301-47	2.46	J301-49	2.46 ± 0.04	0
20	J301-47	2.46	J301-49	2.06 ± 0.04	Border**
20	J301-47	2.46	J301-49	2.86 ± 0.04	Border**

*Input currents may be adjusted within the specified tolerances to obtain the desired indication.

**QUALITY Meter indicates on the border between the GOOD region and one of the BAD regions.

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TABLE XII

Resistance Obtained		Resistance (ohms)
From	To	
J301-5	J301-2	0 to 0.3
J301-32	J301-2	0 to 0.3
J301-38	J301-2	0 to 0.3
J301-40	J301-2	0 to 0 3
J301-42	J301-2	0 to 0.3
J301-44	J301-2	0 to 0 3
J301-46	J301-2	0 to 0.3
J301-48	J301-2	0 to 0.3
J301-50	J301-2	0 to 0 3

3.6.1.2.4.3.2 Resistance Outputs - The bootstrap test circuit shall provide the resistance outputs specified in table XII at connector J301 with the WARNING switch in any position 16 through 20.

3.6 1.2.5 Magnetic Amplifier Module Test Circuit - The magnetic amplifier module test circuit shall meet the following requirements with the MODULE SELECT switch in the MAG AMPL position (7th position).

3.6.1.2.5.1 Resistance Outputs - The Magnetic Amplifier Module test circuit shall provide the resistance outputs listed in table XIII.

3.6.1.2.5.2 QUALITY Meter Indications - The Magnetic Amplifier Module test circuit shall be capable of causing the QUALITY meter to indicate on the border marks between the GOOD region and the two BAD regions when ± 5.50 (+1.10, -0.80) microamperes is applied to the pins specified in table XIV with the MAG AMPL switch in the designated position. Pin J1101-24 shall be the "return" for the input current.

3.6 1.2.5 3 Voltage Outputs - The Magnetic Amplifier Module test circuit shall provide +10 000 ± 0.015 volts d-c output between connector J1101-22 and J1101-24 (ground), and 115 ± 0.6, 119.5 ± 1.6, and 107.5 ± 1.6 volts a-c between connector J1101-23 and J1101-24 with the AC VOLTAGE SELECT switch set at NOM , HIGH and LOW, respectively

3 6 1.2.6 Magnetic Integrator Module Test Circuit - The magnetic integrator module test circuit shall meet the following requirements with the MODULE SELECT switch in the INT position (8th position).

3.6.1.2.6.1 Voltage Outputs - The Magnetic Integrator Module test circuit shall provide the voltage outputs at connector J701 as listed in table XV.

3 6 1.2.6 2 Preintegrate Conditions - The Magnetic Integrator Module test circuit shall provide the conditions listed in table XVI prior to placing the INTEGRATE switch in the START position.

3.6 1.2.6.3 Integrate Conditions - The Magnetic Integrator Module test circuit shall provide the following conditions and those listed in table XVII when the INTEGRATE switch is set to START and 27 ± 2 volts d-c is applied between connector J701-8 and J701-3 (ground). The INTEGRATE TIME meter shall start when J701-7 is shorted to J701-13 and shall stop when J701-9 is shorted to J701-12. A timer accurate to ± 0.01 percent shall be used to check against the time recorded on the INTEGRATE TIME meter. The times shall compare within ± 0.2 percent of the indicated reading ± 0.035 seconds for any time interval up to 30 seconds.

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TABLE XIII

MAG AMPL Switch Position	Resistance Obtained		Resistance (ohms)
	From	To	
all	J1001-3	J1001-24	1 K ± 52.4*
all	J1001-7	J1001-24	1 K ± 52.4*
all	J1001-11	J1001-24	1 K ± 52.4*
all	J1001-15	J1001-24	1 K ± 52.4*
all	J1001-19	J1001-24	1 K ± 52.4*
1	J1001-22	J1001-1	10 K ± 114
2	J1001-22	J1001-1	2 K ± 22.7
3	J1001-22	J1001-1	1 K ± 11.4
4	J1001-22	J1001-1	680 ± 9.13
5	J1001-22	J1001-5	10 K ± 114
6	J1001-22	J1001-5	2 K ± 22.7
7	J1001-22	J1001-5	1 K ± 11 4
8	J1001-22	J1001-5	680 ± 9.13
9	J1001-22	J1001-9	10 K ± 114
10	J1001-22	J1001-9	2 K ± 22.7
11	J1001-22	J1001-9	1 K ± 11.4
12	J1001-22	J1001-9	680 ± 9.13
13	J1001-22	J1001-13	10 K ± 114
14	J1001-22	J1001-13	2 K ± 22 7
15	J1001-22	J1001-13	1 K ± 11 4
16	J1001-22	J1001-13	680 ± 9.13
17	J1001-22	J1001-17	10 K ± 114
18	J1001-22	J1001-17	2 K ± 22.7
19	J1001-22	J1001-17	1 K ± 11.4
20	J1001-22	J1001-17	680 ± 9.13
all	J1001-1	J1001-4	0 to 0.3
all	J1001-5	J1001-8	0 to 0.3
all	J1001-9	J1001-12	0 to 0.3
all	J1001-13	J1001-16	0 to 0.3
all	J1001-17	J1001-20	0 to 0.3
all	J1001-21	J1001-24	0 to 0.3

*These resistors shall be capable of dissipating up to 2.5 watts at +71°C.

TABLE XIV

MAG AMPL Switch Position	Input Current Applied
1-4	J1101-2
5-8	J1101-6
9-12	J1101-10
13-16	J1101-14
17-20	J1101-18

3.6.1.2.7 Phi Servo Module Test Circuit - The phi servo module test circuit shall meet the following requirements when the MODULE SELECT switch is in the SVO position (9th position).

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TABLE XV

Voltage Obtained Between J701-3 (Gnd) and	Voltage
J701-10	+10.000 ± 0.015 v d-c
J701-11	-10.000 ± 0.015 v d-c
J701-14	115 ± 0.8 v a-c (AC VOLTAGE SELECT switch set at NOM.)
J701-14	107.5 ± 1.6 v a-c (AC VOLTAGE SELECT switch set at LOW)
J701-14	119.5 ± 1.6 v a-c (AC VOLTAGE SELECT switch set at HIGH)
J701-18	27.0 ± 2.0 v d-c

TABLE XVI

Condition Obtained		Conditions
From	To	
J701-1	J701-3	-10.000 ± 0.015 v d-c
J701-6	J701-1	820 ± 11 ohms
J701-8	J701-3	Resistance greater than 10 megohms
J701-5	J701-6	Resistance greater than 10 megohms
J701-2	J701-3	Resistance greater than 10 megohms
J701-4	J701-3	-10.000 ± 0.015 v d-c with INTEGRATOR switch in position 2 or 4
J701-4	J701-3	+2.000 ± 0.010 v d-c with INTEGRATOR switch in position 1 or 3

TABLE XVII

Condition Obtained		Conditions
From	To	
J701-1	J701-3	Resistance greater than 10 megohms
J701-2	J701-3	4.000 ± 0.017 v d-c with J701-7 shorted to J701-12 and INTEGRATOR switch set to 1 or 2
J701-2	J701-3	0.75 ± 0.008 v d-c with J701-7 shorted to J701-12 and INTEGRATOR switch set to 3 or 4
J701-4	J701-3	Resistance greater than 10 megohms with J701-7 shorted to J701-12
J701-5	J701-6	0 to 0.30 ohm

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3.6.1.2.7.1 Synchro Outputs - The Phi Servo Module test circuit shall provide synchro outputs of -60, -40, -20, +20, +40 and +60 degrees within an accuracy of ± 1 minute with the SERVO test switch in positions 1 through 6 respectively. The synchro outputs shall be obtained at connector J401 as follows: J401-12(S1), J401-11 (S2), and J401-4 (S3). The connection for S1 and S3 shall be reversed when RELAY-LIN-COS switch is in RELAY position.

3.6.1.2.7.2 Potentiometer Positions Tests - The Phi Servo Module test circuit shall test the positions of the linear and cosine potentiometers in positions 1 through 6 of the SERVO switch. The test circuit for the linear and cosine potentiometer tests shall meet the following requirements.

3.6.1.2.7.2.1 Linear Potentiometer Position Tests - QUALITY Meter Indications - The test circuit for the linear potentiometer position tests shall cause the QUALITY meter to indicate as specified in table XVIII with the applied input voltages connected between connector pins J401-3 and J401-8 (ground) with the RELAY-LIN-COS switch set to LIN or RELAY positions. The "QUALITY Meter 0 Mark" column is for reference and shall not be a test requirement.

3.6.1.2.7.2.2 Cosine Potentiometer Position Tests - QUALITY Meter Indications - The test circuit for the cosine potentiometer position tests shall cause the QUALITY meter to indicate as specified in table XIX with the applied input voltages connected between connector pins J401-14 and J401-8 (ground) with the RELAY-LIN-COS switch set to COS position. The QUALITY Meter 0 Mark" column is for reference and shall not be a test requirement.

TABLE XVIII

SERVO Switch Position	Input Voltage (volts d-c)		
	QUALITY Meter 0 Mark	QUALITY Meter Border Marks Between the GOOD Region and Two BAD Regions	
1	6.667 ± 0.010	6.737 ± 0.015	6.597 ± 0.015
2	4.444 ± 0.010	4.514 ± 0.015	4.374 ± 0.015
3	2.222 ± 0.010	2.292 ± 0.015	2.152 ± 0.015
4	-2.222 ± 0.010	-2.292 ± 0.015	-2.152 ± 0.015
5	-4.444 ± 0.010	-4.514 ± 0.015	-4.374 ± 0.015
6	-6.667 ± 0.010	-6.737 ± 0.015	-6.597 ± 0.015

TABLE XIX

SERVO Switch Position	Input Voltage (volts d-c)		
	QUALITY Meter 0 Mark	QUALITY Meter Border Marks Between the GOOD Region and Two BAD Regions	
1	5.000 ± 0.020	5.075 ± 0.020	4.925 ± 0.020
2	7.660 ± 0.020	7.735 ± 0.020	7.585 ± 0.020
3	9.397 ± 0.020	9.472 ± 0.020	9.322 ± 0.020
4	9.397 ± 0.020	9.472 ± 0.020	9.322 ± 0.020
5	7.660 ± 0.020	7.735 ± 0.020	7.585 ± 0.020
6	5.000 ± 0.020	5.075 ± 0.020	4.925 ± 0.020

3.6.1.2.7.3 Potentiometer Impedance - The Phi Servo Module test circuit shall

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test the impedance of the cosine potentiometer in position 7 of the SERVO switch. The QUALITY meter shall indicate as specified in table XX with the applied resistance connected between connector J401-8 and J401-10 and the RELAY-LIN-COS switch set to COS position. The resistance for causing the QUALITY meter to indicate 0 is for reference and shall not be a test requirement.

TABLE XX

Applied Resistance (ohms)	QUALITY Meter Indication
500.00 ± 0.50	0 mark
502.00 ± 0.50	Right side GOOD/BAD border mark
498.0 ± 0.50	Left side GOOD/BAD border mark

3.6.1.2.7.4 Voltage Outputs - The Phi Servo Module test circuit shall provide the voltage outputs listed in table XXI at connector J401.

TABLE XXI

Voltage Obtained From J401-9 (gnd) To	Voltage
J401-15	-10.000 ± 0.015 V DC
J401-7	+10.000 ± 0.015 V DC
J401-10	+10.000 ± 0.015 V DC
J401-6	27.0 ± 2.0 V DC
J401-2	27.0 ± 2.0 V DC with RELAY-LIN-COS switch set to RELAY
J401-1	115.0 ± 0.8 V AC with AC VOLTAGE SELECT switch set at NOM.
J401-1	107.5 ± 1.6 V AC with AC VOLTAGE SELECT switch set at LOW
J401-1	119.5 ± 1.6 V AC with AC VOLTAGE SELECT switch set at HIGH

3.6.1.2.8 Shaping Network Module Test Circuit - The shaping network module test circuit shall perform as indicated below with the MODULE SELECT switch in the NTWK position (10th position).

3.6.1.2.8.1 Output Currents - The Shaping Network Module test circuit shall provide the output currents listed in table XXII at connector J1101 through the entire specified resistance load ranges.

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TABLE XXII

NETWORK Switch Position	TEST POINT Switch Position	Output Current: Measured Through a Resistor Connected From J1101-50 (gnd) To	Output Current: D-C (ma)	Resistance Load Range (ohms)
1	1	J1101-1	3.02 ± 0.02	1 K to 2 K
1	2	J1101-1	9.94 ± 0.05	860 to 1 K
1	3	J1101-1	11.82 ± 0.06	750 to 1 K
2	1	J1101-3	3.02 ± 0.02	1 K to 2 K
2	2	J1101-3	9.94 ± 0.05	860 to 1 K
2	3	J1101-3	11.82 ± 0.06	750 to 1 K
3	1	J1101-5	3.02 ± 0.02	1 K to 2 K
3	2	J1101-5	9.94 ± 0.05	860 to 1 K
3	3	J1101-5	11.82 ± 0.06	750 to 1 K
4	1	J1101-7	3.02 ± 0.02	1 K to 2 K
4	2	J1101-7	9.94 ± 0.05	860 to 1 K
4	3	J1101-7	11.82 ± 0.06	750 to 1 K
5	1	J1101-9	3 02 ± 0.02	1 K to 2 K
5	2	J1101-9	9 94 ± 0.05	860 to 1 K
5	3	J1101-9	11.82 ± 0.06	750 to 1 K
6	1	J1101-11	3.02 ± 0.02	1 K to 2 K
6	2	J1101-11	9.94 ± 0.05	860 to 1 K
6	3	J1101-11	11.82 ± 0.06	750 to 1 K
7	1	J1101-13	3.02 ± 0.02	1 K to 2 K
7	2	J1101-13	9.94 ± 0.05	860 to 1 K
7	3	J1101-13	11.82 ± 0.06	750 to 1 K
8	1	J1101-15	-3.02 ± 0.02	1 K to 2 K
8	2	J1101-15	-9.94 ± 0.05	860 to 1 K
8	3	J1101-15	-11.82 ± 0.06	750 to 1 K
9	1	J1101-17	-3.02 ± 0.02	1 K to 2 K
9	2	J1101-17	-9.94 ± 0.05	860 to 1 K
9	3	J1101-17	-11.82 ± 0.06	750 to 1 K
10	1	J1101-19	-3.02 ± 0.02	1 K to 2 K
10	2	J1101-19	-9.94 ± 0.05	860 to 1 K
10	3	J1101-19	-11.82 ± 0.06	750 to 1 K
11	1	J1101-21	-3.02 ± 0.02	1 K to 2 K
11	2	J1101-21	-9.94 ± 0.05	860 to 1 K
11	3	J1101-21	-11.82 ± 0.06	750 to 1 K
12	1	J1101-23	-3.02 ± 0.02	1 K to 2 K
12	2	J1101-23	-9.94 ± 0.05	860 to 1 K
12	3	J1101-23	-11.82 ± 0.06	750 to 1 K
13	1	J1101-25	-3.02 ± 0.02	1 K to 2 K
13	2	J1101-25	-9.94 ± 0.05	860 to 1 K
13	3	J1101-25	-11.82 ± 0.06	750 to 1 K
14	1	J1101-27	-3.02 ± 0.02	1 K to 2 K
14	2	J1101-27	-9.94 ± 0.05	860 to 1 K
14	3	J1101-27	-11.82 ± 0.06	750 to 1 K
15	1	J1101-29	-3.02 ± 0.02	1 K to 2 K
15	2	J1101-29	-9.94 ± 0.05	860 to 1 K
15	3	J1101-29	-11.82 ± 0.06	750 to 1 K

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TABLE XXII (CONT'D)

NETWORK Switch Position	TEST POINT Switch Position	Output Current: Measured Through a Resistor Connected From J1101-50 (gnd) To	Output Current: D-C (ma)	Resistance Load Range (ohms)
16	1	J1101-31	-3.02 ± 0.02	1 K to 2 K
16	2	J1101-31	-9.94 ± 0.05	860 to 1 K
16	3	J1101-31	-11.82 ± 0.06	750 to 1 K
17	1	J1101-18	6.50 ± 0.03	900 to 1 K
17	2	J1101-18	13.400 ± 0.065	720 to 1 K
17	3	J1101-18	14.20 ± 0.07	700 to 1 K
18	1	J1101-35	-6.50 ± 0.03	900 to 1 K
18	2	J1101-35	-13.400 ± 0.065	720 to 1 K
18	3	J1101-35	-14.20 ± 0.07	700 to 1 K
19	1	J1101-37	-5.30 ± 0.03	830 to 1 K
19	2	J1101-37	-11.40 ± 0.06	730 to 1 K
19	3	J1101-37	-13.500 ± 0.065	700 to 1 K
20	1	J1101-41*	-2.800 ± 0.025	1 K to 2.1 K
20	2	J1101-41*	-8.000 ± 0.065	1 K to 1 2 K
20	3	J1101-41*	-13.00 ± 0.10	830 to 1 K

*J1101-39 shall be shorted to J1101-40.

3.6.1 2.8.2 QUALITY Meter Indications - The Shaping Network Module test circuit shall cause the QUALITY meter to indicate as specified in table XXIII with the applied input currents. The "QUALITY Meter 0 Mark" column in table XXIII is for reference and shall not be a testing requirement. The tolerances on the values obtained in this paragraph shall be as determined in the next paragraph.

3.6.1.2 8.3 Tolerances - There are no fixed tolerances on the values in table XXIII Instead the total Shaping Network test circuit error shall be determined by using the values obtained in 3.6.1.2.8.1 and 3 6.1.2.8.2. The total circuit error shall be computed as follows for each TEST POINT switch position at each NETWORK switch position

(1) The value obtained per 3 6 1.2.8 1 shall be I_{in}. Solve for I_o in the applicable equation of table XXIV

(2) Determine the difference between I_o and the value in the "QUALITY Meter 0 Mark" of table XXIII. If I_o is less than the value in table XXI, the difference is negative. Let this difference value be "Error 1".

(3) Determine the difference between the value obtained per 3 6.1.2.8 2 for the right border mark and the value specified in the "Right Border Mark" column of table XXIII If the value obtained per 3.6 1.2.8 2 is less than the value specified in table XXIII, the difference value is negative. Let this difference value be "Error 2".

(4) Determine the difference between the value obtained per 3.6.1.2.8.2 for left border mark and the value specified in the "Left Border Mark" column of table XXIII. If the value obtained per 3.6.1.2.8.2 is less than the value specified in table XXIII, the difference is negative. Let this difference value be "Error 3".

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TABLE XXIII

NETWORK Switch Position	TEST POINT Switch Position	Input Current (ma)			
		Input Current Applied To	QUALITY Meter 0 Mark	QUALITY Meter on Border Marks Between the GOOD Region and Two BAD Regions	
				Right Border Mark	Left Border Mark
1	1	J1101-2	2.455	2 480	2.430
1	2	J1101-2	4 081	4.106	4.056
1	3	J1101-2	4.317	4.342	4.292
2	1	J1101-4	2.455	2.480	2 430
2	2	J1101-4	4.081	4.106	4.056
2	3	J1101-4	4.317	4.342	4.292
3	1	J1101-6	2.455	2.480	2.430
3	2	J1101-6	4.081	4.106	4.056
3	3	J1101-6	4.317	4.342	4.292
4	1	J1101-8	2 455	2.480	2.430
4	2	J1101-8	4.081	4.106	4.056
4	3	J1101-8	4.317	4.342	4.292
5	1	J1101-10	2.455	2 480	2.430
5	2	J1101-10	4.081	4.106	4.056
5	3	J1101-10	4.317	4.342	4.292
6	1	J1101-12	2.455	2.480	2.430
6	2	J1101-12	4.081	4.106	4.056
6	3	J1101-12	4.317	4 342	4.292
7	1	J1101-14	2.455	2.480	2.430
7	2	J1101-14	4.081	4.106	4.056
7	3	J1101-14	4.317	4.342	4.292
8	1	J1101-16	-2.455	-2.430	-2.480
8	2	J1101-16	-4.081	-4.056	-4.106
8	3	J1101-16	-4.317	-4.292	-4.342
9	1	J1101-33	-2.455	-2.430	-2.480
9	2	J1101-33	-4.081	-4.056	-4.106
9	3	J1101-33	-4.317	-4.292	-4.342
10	1	J1101-20	-2.455	-2.430	-2.480
10	2	J1101-20	-4.081	-4.056	-4 106
10	3	J1101-20	-4.317	-4.292	-4.342
11	1	J1101-22	-2.455	-2.430	-2.480
11	2	J1101-22	-4.081	-4.056	-4.106
11	3	J1101-22	-4.317	-4.292	-4.342
12	1	J1101-24	-2.455	-2 430	-2.480
12	2	J1101-24	-4.081	-4.056	-4.106
12	3	J1101-24	-4.317	-4.292	-4.342
13	1	J1101-26	-2.455	-2.430	-2 480
13	2	J1101-26	-4.081	-4.056	-4 106
13	3	J1101-26	-4 317	-4.292	-4.342
14	1	J1101-28	-2.455	-2.430	-2.480
14	2	J1101-28	-4.081	-4.056	-4.106
14	3	J1101-28	-4.317	-4.292	-4.342
15	1	J1101-30	-2.455	-2.430	-2.480
15	2	J1101-30	-4.081	-4.056	-4.106
15	3	J1101-30	-4.317	-4.292	-4.342

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TABLE XXIII (CONT'D)

NETWORK Switch Position	TEST POINT Switch Position	Input Current (ma)			
		Input Current Applied To	QUALITY Meter 0 Mark	QUALITY Meter on Border Marks Between the GOOD Region and Two BAD Regions	
				Right Border Mark	Left Border Mark
16	1	J1101-32	-2.455	-2.430	-2.480
16	2	J1101-32	-4.081	-4.056	-4.106
16	3	J1101-32	-4.317	-4.292	-4.342
17	1	J1101-34	6.035	6.073	5.997
17	2	J1101-34	10.835	10.905	10.765
17	3	J1101-34	11.137	11.217	11.057
18	1	J1101-36	-6.035	-5.997	-6.073
18	2	J1101-36	-10.835	-10.765	-10 905
18	3	J1101-36	-11.137	-11.057	-11 217
19	1	J1101-38	-5.245	-5.191	-5 299
19	2	J1101-38	-10.761	-10.675	-10.847
19	3	J1101-38	-12.432	-12.332	-12.532
20	1	J1101-42*	-2.396	-2.364	-2.428
20	2	J1101-42*	-4.095	-4.068	-4.122
20	3	J1101-42*	-5.409	-5.364	-5.454

*J1101-39 shall be shorted to J1101-40 and a 1000 ± 50 ohm resistor shall be connected between J1101-41 and J1101-50.

TABLE XXIV

NETWORK Switch Position for Which Error Is Being Computed	Equation
1 through 16	$I_o = 1.365 \text{ Ln } (2 I_{in})$
17 or 18	$I_o = I_{in} + 2.73 \text{ Ln Cos } (5 I_{in})$
19	$I_o = 19\ 144 \text{ Sin } (3 I_{in})$
20	$I_o = 4.34064 - 1\ 4 \text{ Ln Tan } (5 I_{in})$

(5) Calculate "Error 4" by taking the algebraic difference between "Error 1" and "Error 2".

(6) Calculate "Error 5" by taking the algebraic difference between "Error 1" and "Error 3".

(7) The absolute value of both the "Error 4" and "Error 5" values shall be equal to or less than the applicable value specified in table XXV.

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TABLE XXV

NETWORK Switch Position	TEST POINT Switch Position	Maximum "Error 4" or "Error 5" Value (ma)
1 through 16	1, 2 or 3	0.014
17 or 18	1	0.022
17 or 18	2	0.030
17 or 18	3	0.035
19	1	0.032
19	2	0.039
19	3	0.045
20	1	0.016
20	2	0.021
20	3	0.026

3 6.1.2.8.4 Voltage Outputs - The Shaping Network Module test circuit shall provide +10.000 ± 0.015 volts d-c from connector J1101-45 and J1101-46 to J1101-50 (ground), and -10.000 ± 0.015 volts d-c from J1101-47 and J1101-48 to J1101-50 (ground).

3.6.1.2.9 Adapter Module Test Circuit - The adapter module test circuit shall meet the following requirements with the MODULE SELECT switch in the ADPT position (11th position).

3.6.1.2.9.1 Requirements - The adapter module test circuit shall cause the QUALITY meter to indicate as specified below with the applied resistance values connected to connectors J601 and J603.

(1) With ADAPTER switch in position 1, a 1000 ± 0.05 ohms resistance shall be connected between J601-1 and J601-50 and a 1000 ohm adjustable resistance connected between J603-1 and J603-50. QUALITY meter shall indicate on 0 mark when resistance connected to J603 is adjusted to 1000 ± 2.1 ohms. (This shall not be a test requirement.) QUALITY meter shall indicate on left border mark between GOOD and BAD region when resistance connected to J603 is adjusted to 990 ± 2.1 ohms. QUALITY meter shall indicate on right border mark between GOOD and BAD regions when resistance connected to J603 is adjusted to 1010 ± 2.1 ohms.

(2) (1) above shall be repeated with ADAPTER switch in positions 2 through 49 and resistances connected to J601 and J603 pins corresponding to ADAPTER switch position.

NOTE J601-50 and J603-50 are used throughout this test as ground connections.

3.6.1.2.9.2 Polarity of Resistance Measurements - In all positions except 22, 26, and 32, one end of the resistors being compared shall be at a positive potential and the other end at a ground potential. In positions 22, 26, and 32, one end of the resistors being compared shall be at a negative potential and the other end at a ground potential.

3.6.1.2.10 Switch Module Test Circuit - The switch module test circuit shall meet

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the following requirements with the MODULE SELECT switch in the SW position (12th position).

3.6.1.2.10.1 Stik Time and Ejection Velocity Resistor Tests - QUALITY Meter Indications - The Switch Module test circuit shall test the stik time and ejection velocity resistors in positions 1 through 10 of the SWITCH switch. The test circuit shall cause the QUALITY meter to indicate as specified in tables XXVI and XXVII with the applied resistance values connected to connector J501. The "QUALITY Meter 0 Mark" columns are for reference and shall not be a test requirement.

TABLE XXVI

SWITCH Switch Position	Ejection Velocity Resistor (ohms)	Stik Time Resistor (ohms)*		
		QUALITY Meter 0 Mark	QUALITY Meter Border Marks Between the GOOD Region and Two BAD Regions	
			Right Border Mark	Left Border Mark
	Jumper connected between J501-6 and J501-7· resistor connected from J501-5 to J501-7.	Jumper connected between J501-3 and J501-4, resistor connected from J501-4 to J501-13		
1	000	94,520 ± 500	103,972 ± 5000	85,068 ± 2500
2	47,100 ± 24.0	55,900 ± 300	61,490 ± 1850	50.310 ± 1500
3	23,550 ± 12.0	34,190 ± 170	37,609 ± 1130	30,771 ± 1000
4	15,670 ± 7.8	26,950 ± 135	29,645 ± 890	24,255 ± 730
5	11,770 ± 6.0	20,270 ± 100	22,297 ± 670	18.243 ± 550
6	9,419 ± 4.7	17.300 ± 85	19.030 ± 570	15,570 ± 470
7	7,849 ± 3.9	14,090 ± 70	15.580 ± 470	12,600 ± 380
8	6,728 ± 3.4	12,480 ± 62	13,728 ± 410	11,232 ± 340
9	5,887 ± 2.9	10,590 ± 53	11,649 ± 350	9,531 ± 290
10	5,233 ± 2.6	9,581 ± 48	10,539 ± 320	8,623 ± 260

*Resistors may be adjusted within the specified tolerance to obtain the desired indication.

3.6.1.2.10.2 Set Range and Safe Standoff Resistor Tests - QUALITY Meter Indications - The Switch Module test circuit shall test the set range and safe standoff resistors in the Switch Module in positions 11 through 20 of the SWITCH switch. The test circuit shall cause the QUALITY meter to indicate as specified in tables XXVIII and XXIX with the applied resistance values connected to connector J501. The "QUALITY Meter 0 Mark" columns are for reference and shall not be a test requirement.

3.6.1.2.10.3 Loft Angle Potentiometer Test - QUALITY Meter Indications - The Switch Module test circuit shall test the loft angle potentiometer in position 21 of the SWITCH switch. The test circuit shall be capable of causing the QUALITY meter to indicate as specified below with the applied resistances connected to connector J501.

(1) With the SWITCH switch in position 21, an external voltage source variable within the range of 0 to -10.000 volts shall be connected between J501-10 and J501-11 (ground). For any voltage within this

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voltage range there shall be a corresponding position of the SWITCH MODULE SET potentiometer which shall cause the QUALITY meter to indicate in the GOOD region with the MAX SENS pushbutton depressed. With the QUALITY meter indicating as specified above, the voltage from the variable voltage source shall be equal to the value indicated on the SWITCH MODULE SET dial within ± 0.050 volt d-c.

3.6.1.2.10.4 Loft Run-In and O/S Angle Potentiometer Tests - QUALITY Meter Indications - The Switch Module test circuit shall test the loft run-in and O/S angle potentiometers in positions 22 and 23, respectively, of the SWITCH switch. The test circuit shall be capable of causing the QUALITY meter to indicate as specified below with the voltage applied to connector J501.

TABLE XXVII

SWITCH Switch Position	Ejection Velocity Resistor* (ohms)			Stik Time Resistor (ohms)
	QUALITY Meter 0 Mark	QUALITY Meter Border Marks Between the GOOD Region and Two BAD Regions		
		Left Border Mark	Right Border Mark	
	Jumper connected between J501-6 and J501-7 resistor connected from J501-5 to J501-7.			Jumper connected between J501-3 and J501-4, resistor connected from J501-4 to J501-13.
2	47,100 ± 236	51,810 ± 1550	42,390 ± 1272	55,900 ± 28.0
3	23,550 ± 118	25,900 ± 780	21,200 ± 636	34,190 ± 17.1
4	15,670 ± 78	17,237 ± 517	14,103 ± 423	26,950 ± 13.5
5	11.770 ± 59	12.947 ± 388	10.593 ± 318	20,270 ± 10.1
6	9,419 ± 47	10.360 ± 311	8,478 ± 254	17.300 ± 8.6
7	7,849 ± 39	8.633 ± 259	7,065 ± 212	14,090 ± 7.0
8	6,728 ± 34	7,400 ± 222	6,056 ± 182	12,480 ± 6.2
9	5,887 ± 29	6,475 ± 194	5,300 ± 159	10,590 ± 5.3
10	5,233 ± 26	5,756 ± 173	4,710 ± 141	9,581 ± 4.8

*Resistors may be adjusted within the specified tolerance to obtain the desired indication.

(1) With the SWITCH switch in position 22 an external voltage source variable within the range of 0 to -10.000 volts shall be connected between J501-9 and J501-14 (ground) For any voltage within this voltage range there shall be a corresponding position of the SWITCH MODULE SET potentiometer which shall cause the QUALITY meter to indicate in the GOOD region with the MAX SENS pushbutton depressed. With the QUALITY meter indicating as specified above, the voltage from the variable voltage source shall be equal to the value indicated on the SWITCH MODULE SET dial within ± 0.050 volt d-c.

(2) With the SWITCH switch in position 23 an external voltage source variable within the range of 0 to -10.000 volts shall be connected between J501-8 and J501-14 (ground). For any voltage within this voltage range there shall be a corresponding position of the SWITCH

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MODULE SET potentiometer which shall cause the QUALITY meter to indicate in the GOOD region with the MAX SENS pushbutton depressed. With the QUALITY meter indicating as specified above, the voltage from the variable voltage source shall be equal to the value indicated on the SWITCH MODULE SET dial within ± 0.050 volt d-c.

3.6.1.2.10.5 Voltage Outputs - The Switch Module test circuit shall provide +10.000 ± 0.015 volts d-c between J501-13 and J501-14 (ground); and -10.000 ± 0.015 volts d-c between J501-12 and J501-14 (ground).

TABLE XXVIII

SWITCH Switch Position	Set Range Resistor (ohms)	Safe Standoff Resistor (ohms)*		
		QUALITY Meter 0 Mark	QUALITY Meter Border Marks Between the GOOD Region and Two BAD Regions	
			Right Border Mark	Left Border Mark
	Resistor connected from J501-1 to J501-15.	Resistor connected from J501-2 to J501-13.		
11	616 ± 0.3**	5000 ± 25	5500 ± 165	4500 ± 135
12	1316 ± 0.7	4286 ± 21	4714 ± 140	3857 ± 115
13	1146 ± 0.6	3750 ± 19	4125 ± 125	3385 ± 100
14	1011 ± 0.5	3333 ± 17	3666 ± 110	3000 ± 90
15	901.5 ± 0.5	3000 ± 15	3300 ± 100	2700 ± 81
16	811.9 ± 0.4	2727 ± 14	3000 ± 90	2454 ± 73
17	736.6 ± 0.4	2500 ± 13	2750 ± 83	2250 ± 67
18	672.2 ± 0.4	2308 ± 12	2540 ± 76	2077 ± 62
19	616.2 ± 0.3	2143 ± 11	2357 ± 70	1928 ± 58
20	566.8 ± 0.3	2000 ± 10	2200 ± 66	1800 ± 54

*Resistors may be adjusted within the specified tolerance to obtain the desired indication.

**Resistor shall be connected from J501-1 to J501-25 in lieu of from J501-1 to J501-15.

TABLE XXIX

SWITCH Switch Position		Set Range Resistor (ohms)*		Safe Standoff Resistor (ohms)
	QUALITY Meter 0 Mark	QUALITY Meter Border Marks Between the GOOD Region and Two BAD Regions		
		Left Border Mark	Right Border Mark	
	Resistor Connected from J501-1 to J501-15			Resistor connected from J501-2 to J501-13
11	616 ± 25**	1116 ± 165**	116 ± 60**	5000 ± 2.5
12	1316 ± 6.6	1448 ± 43	1184 ± 36	4286 ± 2.1
13	1146 ± 5.7	1251 ± 38	1031 ± 31	3750 ± 1.9
14	1011 ± 5.1	1112 ± 33	910 ± 27	3333 ± 1.7
15	901.5 ± 4.5	992 ± 30	811 ± 24	3000 ± 1.5

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TABLE XXIX (CONT'D)

SWITCH Switch Position	Set Range Resistor (ohms)*			Safe Standoff Resistor (ohms)
	QUALITY Meter 0 Mark	QUALITY Meter Border Marks Between the GOOD Region and Two BAD Regions		
		Left Border Mark	Right Border Mark	
16	811.9 ± 4.1	893 ± 27	731 ± 22	2727 ± 1.4
17	736.6 ± 3.7	810 ± 24	663 ± 20	2500 ± 1.3
18	672.2 ± 3.4	739 ± 22	605 ± 18	2308 ± 1.2
19	616.2 ± 3.1	678 ± 20	555 ± 17	2143 ± 1.1
20	566.8 ± 2.8	624 ± 19	510 ± 15	2000 ± 1.0

*Resistors may be adjusted within the specified tolerance to obtain the desired indication.
**Resistor shall be connected from J501-1 to J501-25 in lieu of from J501-1 to J501-15.

3.6.1.2.11 Control Module Test Circuit - The control module test circuit shall meet the following requirements with the MODULE SELECT switch in the CM position (13th position).

3.6.1.2.11.1 Burst Height Resistor Tests - QUALITY Meter Indications - The burst height resistor shall be tested in position 1 of the CONTROL MODULE switch. The test circuit shall cause the QUALITY meter to indicate as specified below with the applied resistance connected to connector J801.

(1) The QUALITY meter shall indicate on the 0 mark when 700 ± 0.2 ohms is connected between J801-Z and J801-a; and 1000 ± 1.3 ohms is connected between J801-X and J801-Y. This shall not be a test requirement.

(2) The QUALITY meter shall indicate on the border mark between the GOOD region and left BAD regions when 696.5 ohms is connected between J801-Z and J801-a; and 1000 ± 0.3 ohms is connected between J801-X and J801-Y. The 696.5 ohm resistor may be adjusted ± 1.4 ohms to obtain the desired indication.

(3) The QUALITY meter shall indicate on the border mark between the GOOD region and right BAD region when 700 ± 0.2 ohms is connected between J801-Z and J801-a; and 995 ohms is connected between J801-X and J801-Y. The 995-ohm resistor may be adjusted ± 2.0 ohms to obtain the desired indication.

3.6.1.2.11.2 Target Height 2 and D Value 2 Resistor Test - QUALITY Meter Indications - The target-height-2 and D-value-2 resistors shall be tested in position 2 of the CONTROL MODULE switch. The test circuit shall cause the QUALITY meter to indicate as specified in table XXX with the applied resistances connected to connector J801. The "QUALITY Meter 0 Mark" column is for reference and shall not be a test requirement.

3.6.1.2.11.3 Target Height 1 Resistor Tests - QUALITY Meter Indications - The target-height-1 resistors shall be tested in position 3 through 11 of the CONTROL MODULE switch. The test circuit shall cause the QUALITY meter to indicate as specified in table XXXI with the applied resistances connected to connector J801. The "QUALITY Meter 0 Mark" column is for reference and shall not be a test requirement.

3.6.1.2.11.4 D Value 1 Resistor Test - QUALITY Meter Indications - The D-value-1 resistors shall be tested in position 12 of the CONTROL MODULE switch. The test circuit shall cause the QUALITY meter to indicate as specified below with the applied resistances connected to connector J801.

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TABLE XXX

	QUALITY Meter 0 Mark	QUALITY Meter Border Marks Between the GOOD Region and Two BAD Regions	
		Right Border Mark	Left Border Mark
Resistance Between J801-W and J801-Y (ohms)	Resistance Between J801-Y and J801-V (ohms)	Resistance Between J801-Y and J801-V (ohms)	Resistance Between J801-Y and J801-V (ohms)
200 K ± 100	200 K ± 2000	232 K ± 7000	173 K ± 5200
25 K ± 12.5	25 K ± 200	22.5 K ± 680	27.5 K ± 820

TABLE XXXI

CONTROL MODULE Switch Position	QUALITY Meter 0 Mark	QUALITY Meter Border Marks Between the GOOD Region and Two BAD Regions	
		Left Border Mark	Right Border Mark
	Resistance Between J801-Y and J801-V (ohms)	Resistance Between J801-Y and J801-V (ohms)	Resistance Between J801-Y and J801-V (ohms)
3	20,250 ± 100	22,275 ± 780	18,225 ± 640
4	10,250 ± 51	11.275 ± 400	9,225 ± 320
5	6,917 ± 35	7,609 ± 270	6,225 ± 220
6	5,250 ± 26	5,775 ± 200	4,725 ± 165
7	4,250 ± 21	4,675 ± 165	3,825 ± 135
8	3,583 ± 18	3,941 ± 140	3,225 ± 115
9	3,107 ± 16	3,437 ± 120	2,813 ± 100
10	2,750 ± 14	3,025 ± 105	2,475 ± 86
11	2,472 ± 13	2,719 ± 95	2,225 ± 78

(1) The QUALITY meter shall indicate on the 0 mark when 20,250 ± 100 ohms is connected between J801-Y and J801-V. This shall not be a test requirement.

(2) The QUALITY meter shall indicate on the Left GOOD/BAD border mark when 22,275 ± 780 ohms is connected between J801-Y and J801-V.

(3) The QUALITY meter shall indicate on the Right GOOD/BAD border mark when 18,225 ± 640 ohms is connected between J801-Y and J801-V.

3.6.1.2.11.5 Resistance Output - With the CONTROL MODULE switch in position 12 the resistance between J801-V and J801-W shall be within the range of 0 to 0.3 ohm.

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TABLE XXXII

CONTROL MODULE Switch Position	Condition Obtained		Condition
	From	To	
all	J801-U	J801-H	0 to 0.5 ohm
all	J801-E	J801-L	0 to 0.5 ohm
all	J801-K	J801-C	0 to 0.5 ohm
13	J801-N connected to J801-T		WARN lamp lights
13	J801-P connected to J801-T		WARN lamp lights
13	J801-R connected to J801-T		WARN lamp lights
13	J801-S connected to J801-T		WARN lamp lights
14	J801-B connected to J801-T		WARN lamp lights
14	J801-P connected to J801-T		WARN lamp lights
14	J801-R connected to J801-T		WARN lamp lights
14	J801-S connected to J801-T		WARN lamp lights
14	J801-M	J801-J (gnd)	27 ± 2.0 v d-c
15	J801-M	J801-J (gnd)	27 ± 2.0 v d-c

3.6 1.2.11.6 Attack Mode Tests - Output Conditions - The Control Module test circuit shall test the attack mode circuits in positions 13 through 15 of the CONTROL MODULE switch. The test circuit shall provide the resistance, voltage, and WARN lamp indications specified in table XXXII.

3.6.1.2.11.7 Voltage Outputs - The Control Module test circuit shall provide +10.000 ± 0.015 volts d-c between connector J801-W and J801-J (ground), and -10.000 ± 0.015 volts d-c between J801-V and J801-J (ground).

3.6.2 Case, Test Set CY-4629/ASM-152 - The test set shall be provided with a combination case manufactured in accordance with the applicable drawings of DL 63A24E1.

3.6.2.1 Function - The test set combination case shall be used to house the test set, the Power Cable Assembly, the Special Purpose Cable Assembly, the Module Puller and Disconnect Tool, the hex wrench and the test set handbook.

3.6.3 Cable Assembly, Power, Electrical, Branched CX-10159/ASM-150 - The cable assembly shall be manufactured in accordance with the applicable drawings of DL 63A24E1.

3.6.3.1 Function - The cable assembly shall be used to provide power to the test set.

3.6.4 Cable Assembly, Special Purpose, Electrical CX-11043 ASM-152 - The cable assembly shall be manufactured in accordance with the applicable drawings of DL 63A24E1.

3.6.4.1 Function - The cable assembly shall be used to provide a connection for the Pilot's Control Module to be tested to Connector J801 of the equipment.

3.6.5 Distribution Box J-2556/ASM-152 - The distribution box shall be manufactured in accordance with the applicable drawings of DL 63A24E1.

3.6.5.1 Function - The distribution box shall be used for testing all wiring and and resistors in the MODULE SELECT switch assembly.

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3.6.6 Tool, Module Puller and Disconnect - The module puller and disconnect tool shall be manufactured in accordance with the applicable drawings of DL 63A24E1.

3.6.6.1 Function - The module puller and disconnect tool is used to aid in removal of Computer CP-741 (V)/A modules from the Computer chassis, to mount the modules to the test set and to carry the Computer CP-741(V)/A.

3.6.7 Wrench, Hex - The hex wrench shall be manufactured in accordance with the applicable drawings of DL 63A24E1.

3.6.7.1 Function - The hex wrench shall be used to loosen the set screws in all front panel knobs.

3.7 Workmanship - General workmanship shall be in accordance with MIL-W-27076 and MIL-STD-454, requirement 9. Soldering shall be in accordance with MIL-STD-454, requirement 5.

3.8 Design Data - No data are required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order (see 6.3).

4. QUALITY ASSURANCE PORVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government The Government reserves the right to perform any of the inspections set forth in the Specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Test

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval (see 6.3).

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include the following environmental tests in accordance with the requirements of MIL-T-5422 to the limits specified in

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MIL-T-21200 except as modified by 3.4.6.1 and 3.4.6.2.

(1) Temperature
(2) Altitude
(3) Vibration
(4) Shock
(5) Humidity
(6) Salt Spray
(7) Explosion-proof
(8) Sand and Dust
(9) Fungus

In addition, preproduction tests shall include interference tests in accordance with MIL-I-6181 and drop and water tightness tests in accordance with MIL-T-21200. When required (see 3.4.4), Maintainability Demonstration tests shall be in accordance with MIL-STD-471.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipment shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment (see 6.3).

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

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4.3.4 Reconditioning of Initial Production Test Sample - On completion of the Initial Production Test the equipment shall be reworked by the contractor by replacing all "wear" or damged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests (see 6.3). Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests

(1) Examination of Product
(2) Operational Test
(3) Manufacturing Run-In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation (see 6.3).

4.4.1.3 Manufacturing Run-In Test - Each equipment shall be operated under the conditions specified in this paragraph for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. The direction of vibration should be vertical to the normal mounting plane for 5 minutes

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and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action).

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

*If by contract action the Initial Production Test (see 4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and previous special tests.

(3) Manufacturing run-in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests shall be conducted as required by both the Qualification Phase and Sampling Phase.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in table 5 of MIL-STD-781.

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For the Qualification Phase Test Level C shall be used. The Accept-Reject Criteria for Test Plan VI shall be used.

4.4.3.2 Reliability Sampling Phase Tests - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the Section entitled "Sampling Phase of Production Reliability Tests". For the Sampling Phase Test Level C shall be used. The Accept-Reject criteria for Test Plan VI shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.4.3.2.1 Lot Size for Sampling Phase - The entire contract quantity (with the exception of those equipments used in the Qualification Phase) shall be considered one lot. The test program shall start the first month after the Qualification Phase has been completed. Samples shall be selected and placed on test in the same manner as though one month's production was a lot. The samples shall be tested until an accept or reject decision is reached. In the meantime other equipments constructed shall be shipped. The test results shall be summarized monthly for the procuring activity (see 6.3). The procuring activity reserves the right to stop the acceptance of equipment any time after one or more reject decisions have been reached, pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment quality control, etc., so that the entire contract quantity will show an accept decision.

4.4.3.3 Test Details - Test details such as the length of the Test Cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted by the contractor and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments to check the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows

(1) On an early production equipment, after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance, or special tests, the following action shall be taken

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective

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action intended to reduce the possiblity of the same failure(s) occurring in future tests (see 6.3).

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual tests. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Assurance Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions

Temperature	Room ambient (30° ± 10°C)
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
Input Voltage	115 ± 10% volts a-c (at lowest applicable frequency) 27.0 ± 2.0 volts d-c
Vibration	None

4.5.2 Test Periods - The test may be run continuously or intermittently. However, only the time during which the test set is energized shall count toward the 300-hour test duration. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

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(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test (see 6.3). In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests, and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval (see 6.3). The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all "wear" or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and be resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector (see 6.3).

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 (see 6.3).

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6. NOTES

6. 1 Intended Use - The Test Set, Electronic Circuit Plug-In Unit AN/ASM-152, is intended to be used to determine the operational capability of the plug-in modules of the Weapons Release Computer CP-741(V)/A and the associated Pilot's Control Module (Weapon Control Panel).

6. 2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply

(1) Contract

(2) This Specification

(3) The drawings of DL 63A24E1

(4) Referenced Specifications

6. 3 Ordering Data - Purchasers should exercise any desired options offered herein and procurement documents should specify the following

(1) Title number and date of this specification

(2) Selection of applicable levels of packaging and packing (see 5. 1)

(3) Nonstandard parts and material approval (see 3. 3 2)

(4) Parts replacement data (see 3. 4. 1. 2)

(5) Preproduction test data (see 4. 2. 1)

(6) Accessory material (see 4. 3. 2)

(7) Acceptance test reports (see 4. 4)

(8) Operational test data (see 4. 4. 1. 2)

(9) Reliability test summary (see 4. 4. 3. 2. 1)

(10) Equipment failure (see 4. 4. 5)

(11) Failure report (see 4. 5. 4 1)

(12) Test procedures (see 4. 6)

(13) Rejection and retest data (see 4 9)

6. 4 Definitions -

6. 4. 1 Preferred Parts - Preferred parts are those in either of the following two categories

(1) Parts manufactured to established reliability military specifications having quantitative reliability and qualification approval (QPL) requirements.

(2) Parts which meet the quantitative reliability requirements of the component drawing and quality assurance information requirements of NASC Specification 64A6A100 and are manufactured in a plant listed on a MIL Specification Qualified Products List (QPL) for the generic part type or in a plant which has been approved on the basis of a vendor technical evaluation.

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6.4.2 Preliminary Parts - Preliminary parts are those parts that do not meet the requirements of the Preferred or Qualified category but have been approved for use by the procuring activity on the basis of need, quality, reliability, and vendor information.

6.4.3 Qualified Parts - Qualified parts are those in either of the following two categories

(1) Parts manufactured to conventional "MIL", "MS", or "AN" specifications having Qualification Approval (QPL) requirements.

(2) Parts which meet the acceptance inspection requirements for "Quality Assurance Information" in accordance with NASC Specification 64A7A100 and are manufactured in a plant listed on a Mil Specification Qualified Parts List (QPL) for the generic part type, or in a plant which has been approved on the basis of a vendor technical evaluation.

6.4.4 Vendor Technical Evaluation - A vendor technical evaluation is the obtaining of first-hand knowledge that the manufacturer has the technical capacity to produce products satisfactorily to the quality, reliability, and functional requirements specified The vendor technical evaluation is conducted by and/or under the cognizance of the procuring activity.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-T-81541(AS) TEST SET, ELECTRONIC CIRCUIT PLUG-IN UNIT AN/ASM-152

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

FORM 1426
APR 63

C-8279

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DEPARTMENT OF THE NAVY

Naval Air Systems Command
Washington, D.C. 20360

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